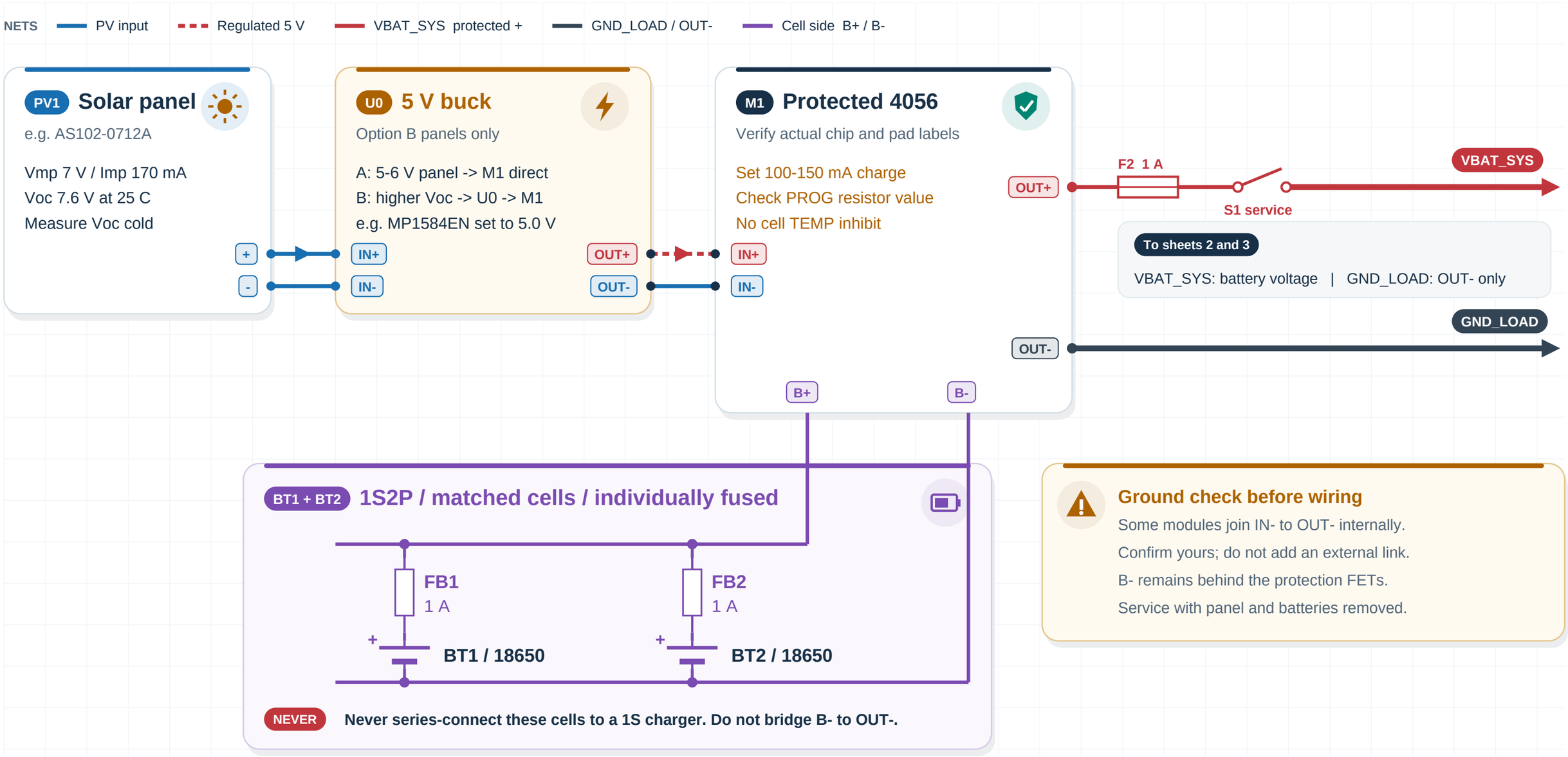
Module A: power, charging & battery

Terminal-level schematic. Amber blocks depend on your panel choice. All loads connect to the protected outputs.



Ground check before wiring

Some modules join IN- to OUT- internally.

Confirm yours; do not add an external link.

B- remains behind the protection FETs.

Service with panel and batteries removed.

RULE Series diodes are not overvoltage protection. The 4056 module has no cell-temperature inhibit: see Known limitations.

Modules B-D: controller, light sensor & low-power LEDs

ATmega328P Pro Mini on its internal 8 MHz clock. D9 drives the low-power LED string directly by PWM through R3; no driver transistor.

NETS — VBAT_SYS protected + — GND_LOAD / OUT- — Logic / sensor / LED drive

VBAT_SYS

GND_LOAD

VCC

GND

M2 BTE13-010A

ATmega328P, verified 8 MHz

Remove regulator + power LED
VCC supplied directly; RAW unused
Fuses: internal 8 MHz, BOD 2.7 V
D8, D10, D11, A0: not used

D7

D9 / PWM

D2

D1 / TX → to Q2 (sheets 3, 6)

A1

LDR1
ambient light

R4
22k

RV1
50k trim

GND_LOAD

SW1 / TEST

GND_LOAD

R3
56R

LED string
low power; + top, - bottom

GND_LOAD

VBAT_SYS

+ **C3 100u / 10 V**

GND_LOAD

VBAT_SYS

+ **C4 100n**

GND_LOAD

Low-power string only

Reference string: 2.56 V drop
I = (VBAT - 2.56) / (R3 + pin R)
Model: 19 / 13 / 10 mA at
4.20 / 3.70 / 3.40 V
Measure it in circuit

Not mains, 12 V or legacy lights
20 mA pin rating is the ceiling

RULE All ground symbols on this sheet mean OUT-. R3 sets the peak current; duty changes the average only. Verify current at 4.20 V and 100% duty.

Module E: telemetry, Tier 2 production (D1 Mini)

Hourly deep sleep on battery power. Same Q2 serial harness as Tier 1 (sheet 6); the D1 Mini is powered from U3.

NETS — VBAT_SYS protected + — ESP 3V3 — GND_LOAD / OUT- — Logic / sensor

VBAT_SYS
GND_LOAD

U3 3.3 V supply

TPS63802 breakout, 3.3 V

VIN: VBAT_SYS
OUT: D1 3V3 pin (Tier 2 only)
Check 3.3 V at 3.0-4.2 V in

3V3

VBAT_SYS
R5
220k
battery sense

M3 D1 Mini

Tier 2: powered from U3 only

5V pin + USB: not connected
Onboard regulator: fitted, unused

A0
3V3
D0
deep-sleep wake
RST
D6
G
GND_LOAD

VBAT_SYS
+
C2 470u / 10 V
GND_LOAD

ESP 3V3
R10
10k

M2 TX
R8 47k
B
C
E
R9
100k
GND_LOAD
Q2 / 2N3904
check E/B/C
GND_LOAD

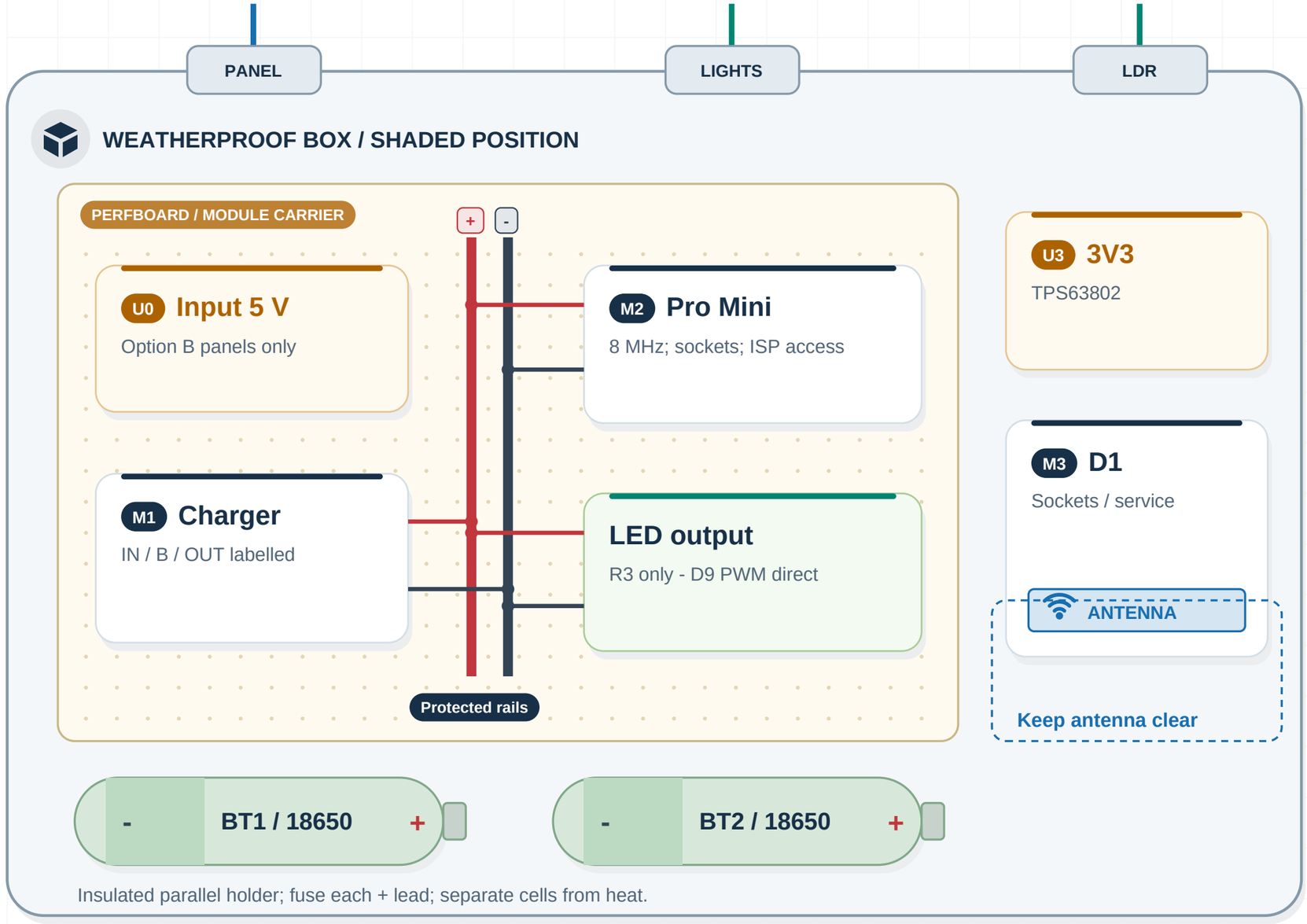
Interface behaviour

Controller (production build) sends a line every ~8 s.
Q2 inverts + level-shifts; YAML declares D6 inverted.
D1 listens up to 20 s per hourly wake, then sleeps.
A0 assumes onboard 220k / 100k divider: calibrate.

SAFETY Disconnect external power AND signal leads before USB/programmer servicing. Never supply the ESP 3V3 pin from the cell directly.

What to assemble inside the enclosure

Top-down placement concept, not a drilling template. Use terminal labels and the wire schedule for electrical connections.



Build in this order

1. Prepare, fuse and flash the Pro Mini.
2. Add R3 (56R) + low-power LED string.
3. Add LDR, RV1 trim + TEST button; check dusk/dawn.
4. Qualify charger (and U0 if fitted).
5. Assemble the protected, fused pack.
6. Add telemetry; commission with Tier 1.

Label every connector

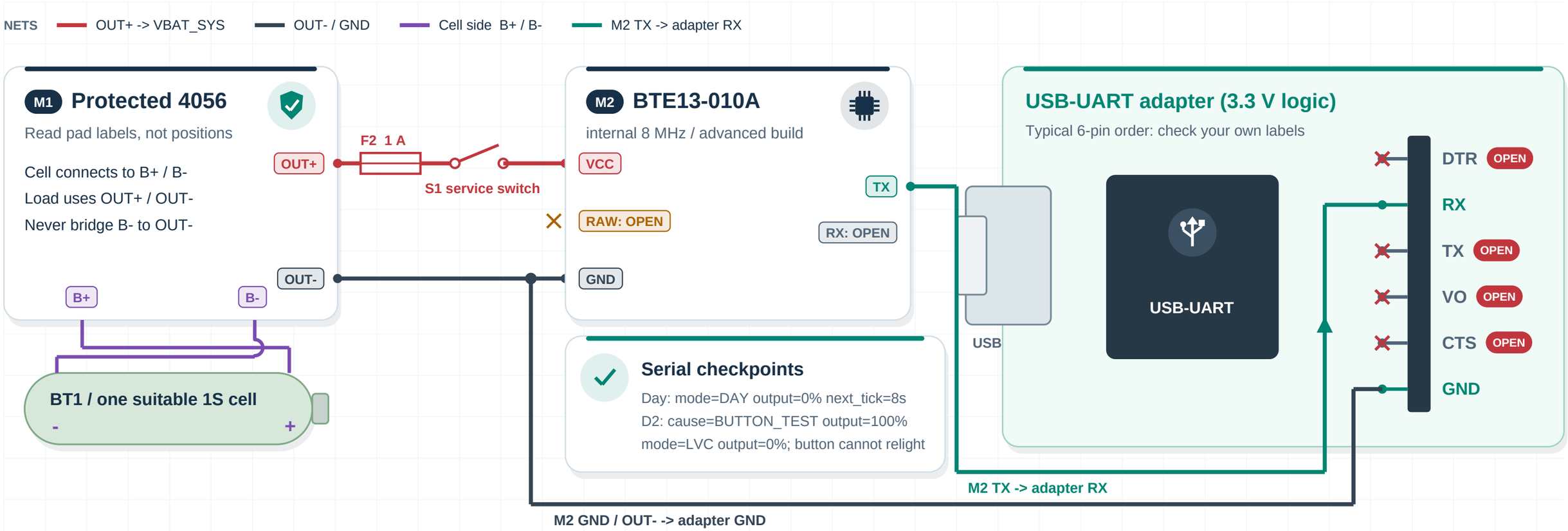
- RED: protected positive / VBAT_SYS
- BLACK: load return / OUT-
- BLUE: panel input pair
- PURPLE: battery-only connections
- GREEN: logic / sensor signals

Before closing the lid

- [ ] Strain relief, polarity, cell insulation.
- [ ] No exposed underside copper.
- [ ] Charger/cell temperature tests passed.
- [ ] 14-day Tier 1 soak test passed.

Staged power-up & wired serial monitor

Optional alternative to Tier 1 telemetry. The USB-UART adapter only listens: connect RX and GND; leave its power and control pins open.



Before closing S1

1. Meter OUT+ to OUT-: polarity correct.
2. Expect approximately the cell voltage.
3. UART power lead already removed.
4. Close S1; compare serial vdd with meter.

Proceed one gate at a time

1	2	3	4	5	6	7
M2	BAT	LDR	LVC	5 V	PV	RUN

USB-UART rules

USB cable powers the adapter itself.
Do not connect VO to M2 VCC.
Leave DTR, TX and CTS open for monitoring.
Keep adapter USB-powered while M2 TX is attached.

RULE SERIAL IS NOT A CHARGE METER: verify panel input, regulated 5 V, battery voltage and charge current with instruments.

Module E: telemetry, Tier 1 Advanced (D1 Mini)

Always-on commissioning instrument: decodes every controller diagnostic line into Home Assistant while you build, calibrate and soak-test.

TIER 1 -> TIER 2 Same wiring. Reflash the controller (production build) and D1 Mini (production profile), then move the D1 supply from USB to U3 (sheet 3).

NETS — UART 9600 baud, inverted — ESP 3V3 — OUT- / GND_LOAD — Battery sense

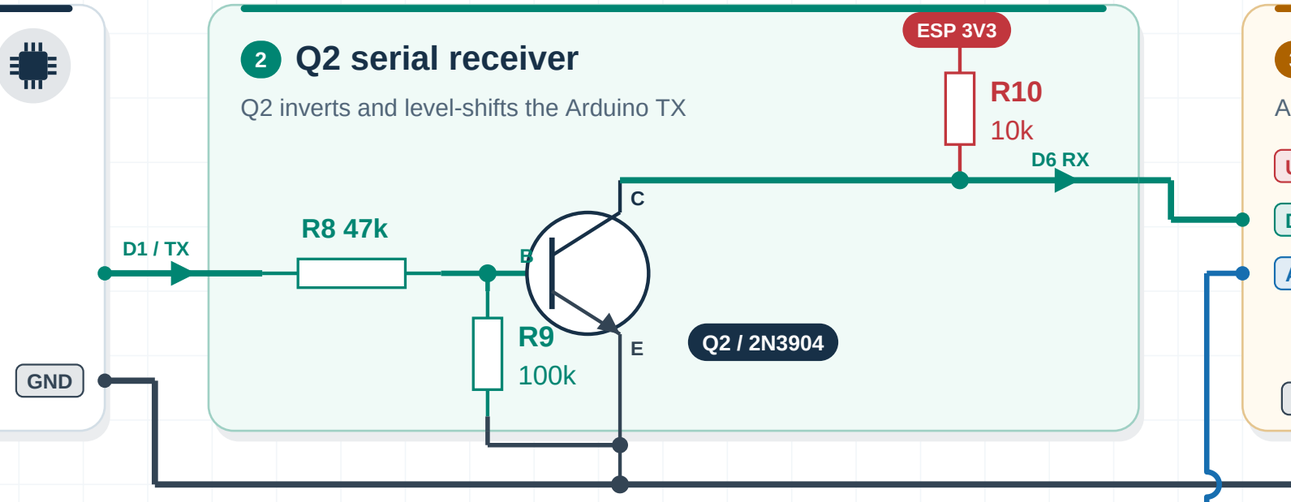
1 M2 Arduino

Flash the advanced build

- VCC — VBAT_SYS
- GND — OUT- / GND_LOAD
- D1/TX — 9600-baud diagnostics

2 Q2 serial receiver

Q2 inverts and level-shifts the Arduino TX



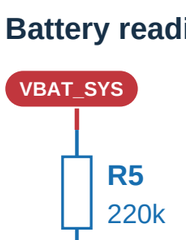
3 M3 Wemos D1 Mini

Always on, powered from USB

- USB — micro-USB power; U3 OUT disconnected
- D6 — inverted UART RX
- A0 — battery divider (R5)
- G

OUT- / GND_LOAD common return only — D1 Mini TX remains OPEN

Battery reading



D1 A0 has its onboard divider too. Calibrate the multiplier against a meter, and compare it with Arduino VDD.

What Home Assistant receives

Arduino: VDD, LDR, sense/mode/night, output duty, LVC/test progress, cause, transition and raw line.

D1: battery voltage/percentage, Wi-Fi, reset, heap and uptime.

Not measured by telemetry

Panel voltage, charge/load current, energy, physical LED current and temperatures: use a meter or power profiler for these.

TIER 1 Tier 1 rules: advanced build on M2 • no USB-UART on M2 • D1 on micro-USB with U3 OUT disconnected • never both supplies on 3V3

System overview

Every module on one sheet. Telemetry reports controller state to Home Assistant; it is not a current or energy meter.

HOW TO USE System-level view. Follow sheets 1, 2, 3 and 6 for terminal-level wiring. The editable KiCad schematic is in hardware/kicad/.

NETS — PV input — VBAT_SYS protected + — GND_LOAD / OUT- — Cell side B+ / B- — UART 9600 baud, inverted — Battery sense — ESP 3V3

PV1 Solar panel
e.g. 1.2 W
PV+ / PV-

U0 5 V input
Option B panels only
Option A: panel direct

M1 Protected charger
TP4056 family + protection
IN+ / IN- B+ / B-

B+
B-

BT1 + BT2 1S2P cells
Each positive lead fused
BATT+ / BATT-

cell-side wiring:
B+ / B- only
loads connect to protected OUT+ only

OUT-
OUT+
F2 1 A
S1 service
OUT+ -> F2 -> S1 -> VBAT_SYS

VBAT_SYS
GND_LOAD
OUT- / GND_LOAD common return

M2 BTE13-010A Pro Mini
Internal 8 MHz, ISP programmed

- VCC <- VBAT_SYS GND <- OUT-
- D7 -> LDR -> A1 -> R4 + RV1 -> OUT-
- D9 -> R3 56R -> LED string -> OUT-
- D1/TX -> R8 -> Q2 base (9600 baud)
- D2 <- SW1 TEST button to OUT-

U3 3.3 V supply
TPS63802 at 3.3 V for M3
VIN <- VBAT_SYS OUT -> M3 (Tier 2)

R8 47k

Q2 2N3904 NPN inverter
B: R8 47k from D1/TX
C: D6 RX; R10 10k -> ESP 3V3
E: OUT-; R9 100k B-E

R5 220k
ESP 3V3
R10
D6 RX

M3 Wemos D1 Mini
Tier 1 always on / Tier 2 hourly

- A0 <- R5 220k <- VBAT_SYS
- 3V3 <- U3 (Tier 2) / USB (Tier 1)
- GND <- OUT-
- D6 <- Q2 collector (inverted RX)
- D0 -> RST (Tier 2 wake)

Home Assistant

Visible in Home Assistant
Arduino: state, VDD, LDR, output, low-voltage progress and raw diagnostics.
D1: battery, Wi-Fi, reset reason, heap and uptime.

Still measured with instruments
Panel/charge voltage and charge/load current or energy.
Physical LED current and temperature.